AgriTrack Data Dictionary

Project: AgriTrack — Web-Based Farm Management System

Purpose: Defines database tables, fields, data types, constraints, and relationships for the backend.

Database Tables and Fields

Table: users

Field	Data type	Description / constraints	Nullable
id	INT	Primary key; auto-increment	No
full_name	VARCHAR(100)	User's full name	No
email	VARCHAR(150)	Unique login email	No
password_hash	VARCHAR(255)	Hashed password; never store plain text	No
role	ENUM('Admin','Manager','Wor ker')	Role for access control	No
is_active	BOOLEAN	Account active status; default TRUE	Yes
created_at	TIMESTAMP	Creation timestamp; default current timestamp	Yes

Table: farms

Field	Data type	Description / constraints	Nullable
id	INT	Primary key; auto-increment	No
name	VARCHAR(150)	Farm name	No
location	VARCHAR(255)	Farm location	No
area_hectares	DECIMAL(10,2)	Farm area in hectares; must be >= 0	No
owner_user_id	INT	Foreign key references users.id	Yes
created_at	TIMESTAMP	Creation timestamp; default current timestamp	Yes

Table: crops

Field	Data type	Description / constraints	Nullable
id	INT	Primary key; auto-increment	No
name	VARCHAR(100)	Crop name	No
variety	VARCHAR(100)	Crop variety	Yes

Field	Data type	Description / constraints	Nullable
description	TEXT	Additional crop details	Yes
created_at	TIMESTAMP	Creation timestamp; default current timestamp	Yes

Table: Planting

Field	Data type	Description / constraints	Nullable
id	INT	Primary key; auto-increment	No
farm_id	INT	Foreign key references farms.id	No
crop_id	INT	Foreign key references crops.id	No
planted_area_hectares	DECIMAL(10,2)	Planted area in hectares; must be >= 0	No
planting_date	DATE	Date the crop was planted	No
expected_harvest_date	DATE	Expected harvest date	Yes
status	VARCHAR(30)	Planting status (e.g., planned, growing, harvested)	No
created_at	TIMESTAMP	Creation timestamp; default current timestamp	Yes

Table: harvests

Field	Data type	Description / constraints	Nullable
id	INT	Primary key; auto-increment	No
planting_id	INT	Foreign key references plantings.id	No
harvest_date	DATE	Date of harvest	No
quantity	DECIMAL(12,2)	Harvested quantity; must be >= 0	No
unit	VARCHAR(20)	Unit such as kg, sacks, or crates	No
notes	TEXT	Optional harvest notes	Yes
recorded_by	INT	Foreign key references users.id	No
created_at	TIMESTAMP	Creation timestamp; default current timestamp	Yes

Table: inventory

Field	Data type	Description / constraints	Nullable
id	INT	Primary key; auto-increment	No
farm_id	INT	Foreign key references farms.id	No
harvest_id	INT	Foreign key references harvests.id; harvested products only	No

Field	Data type	Description / constraints	Nullable
item_name	VARCHAR(150)	Name of harvested product	No
quantity	DECIMAL(12,2)	Current quantity; must be >= 0	No
unit	VARCHAR(20)	Unit such as kg, sacks, or crates	No
updated_at	TIMESTAMP	Last update timestamp; default current timestamp	Yes

Table Relationship

Relationship	Meaning
users 1:M farms	A user may own/manage multiple farms through farms.owner_user_id.
users 1:M harvests	A user may record many harvests through harvests.recorded_by.
farms 1:M plantings	Each planting belongs to one farm.
crops 1:M plantings	A crop type may be used in many planting records.
plantings 1:M harvests	One planting may have multiple harvest records.
farms 1:M inventory	Each inventory record belongs to one farm.
harvests 1:M inventory	Each inventory record references one harvest; a harvest may be linked to multiple inventory records.